

Computer vision project

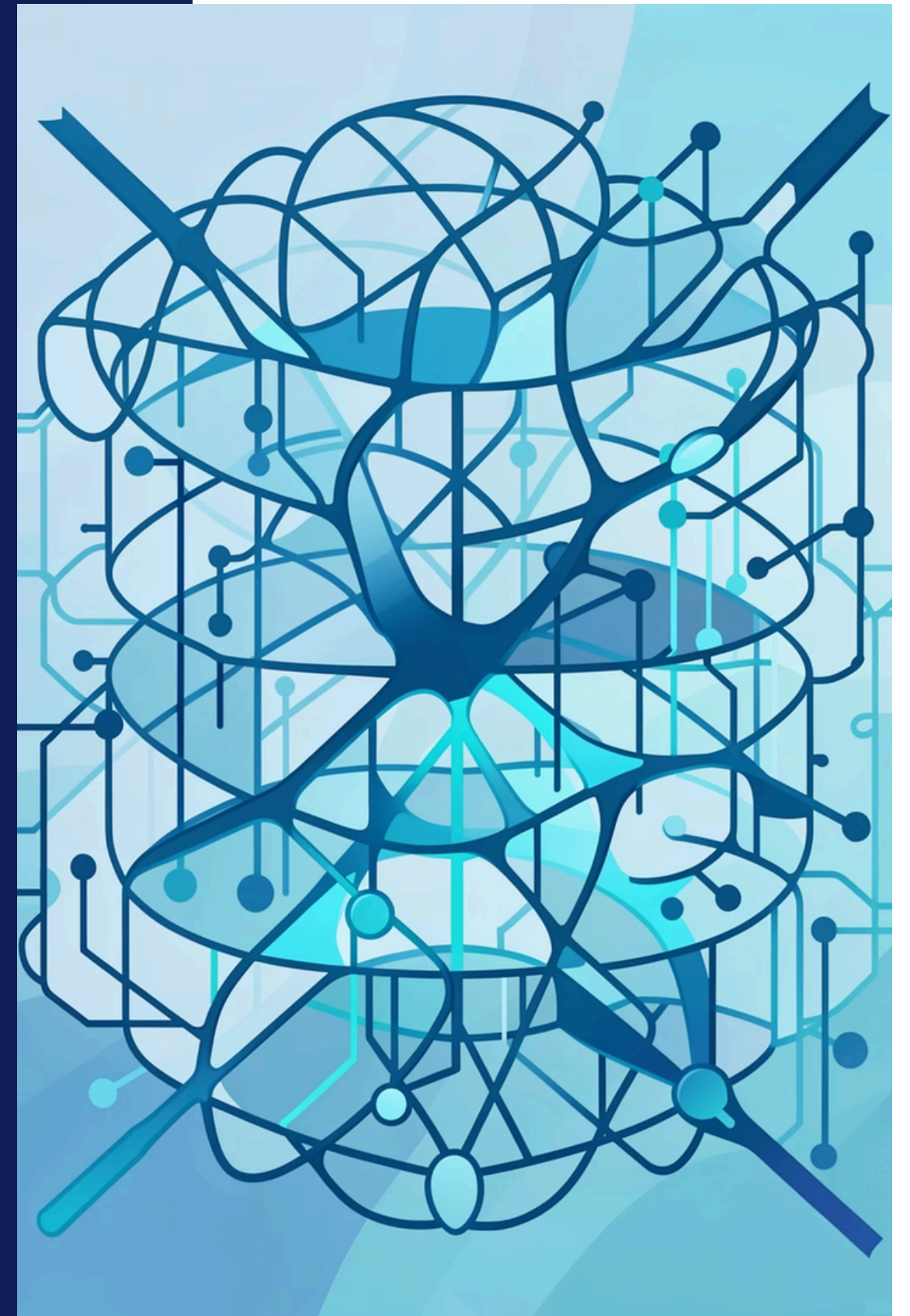
Presented By The Fantastic Four:

Eya Cherif

Suzana Cracco

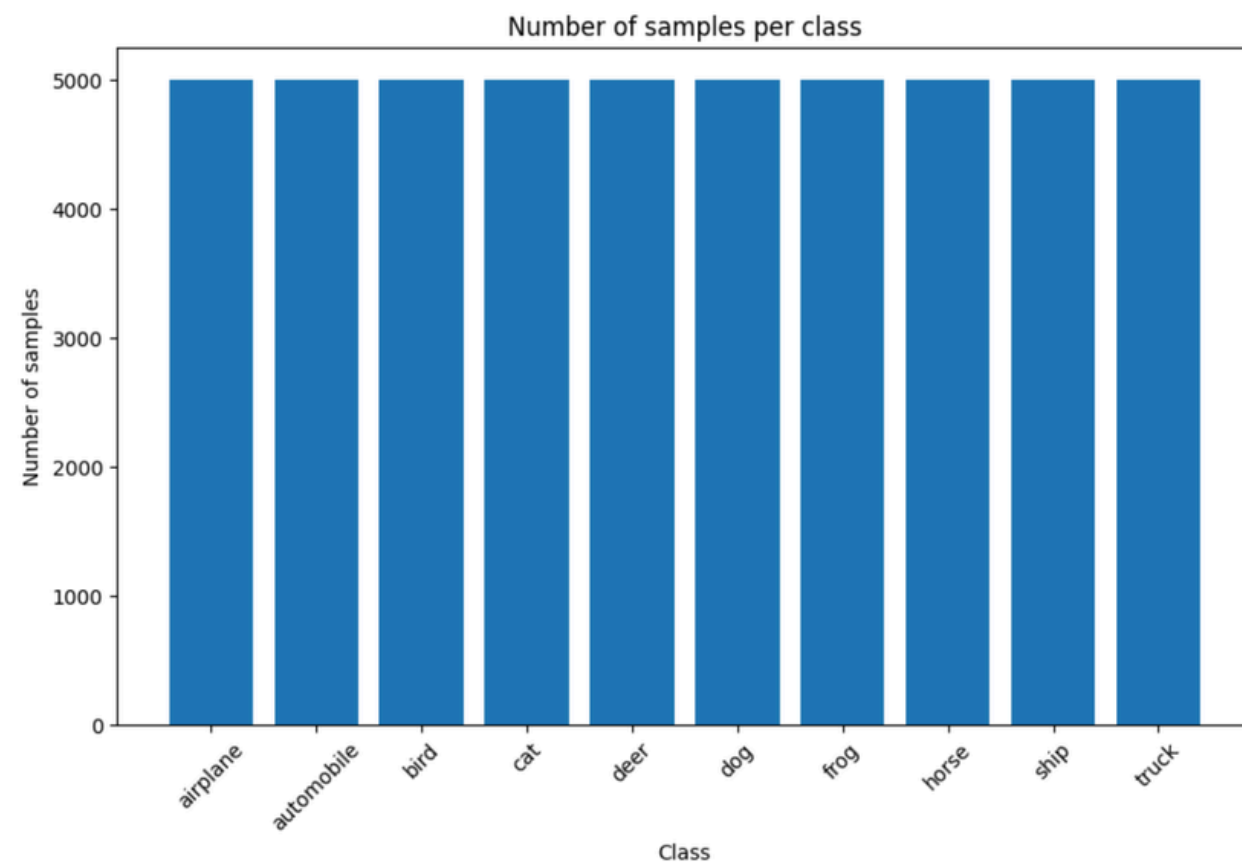
Aitor Vergara

Alejandra Cárdenas



DATASET: CIFAR10

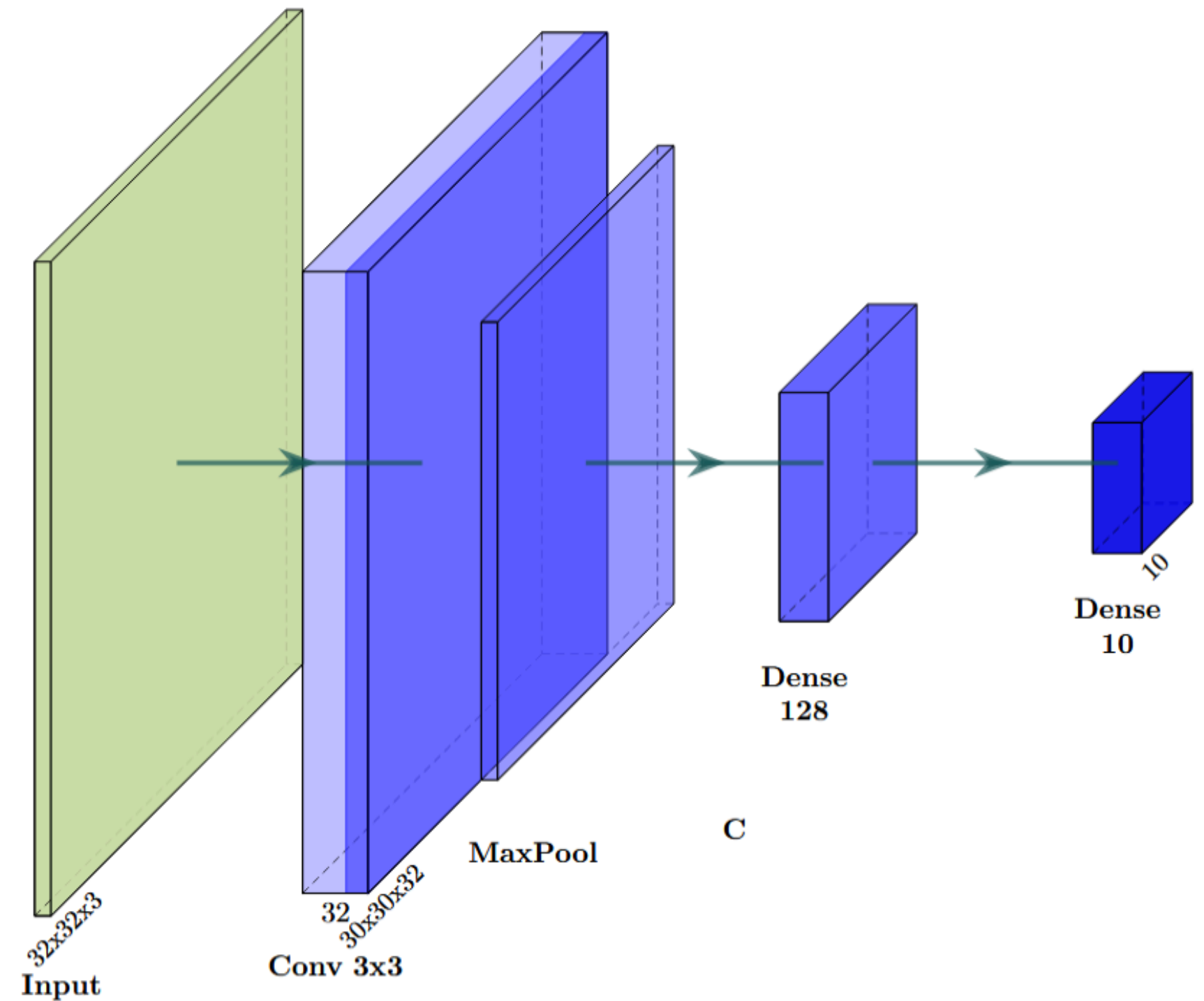
- 60000 images
- 10 classes
- 32x32 pixels



1-CNN-LAYERED MODEL

Hyperparameters:

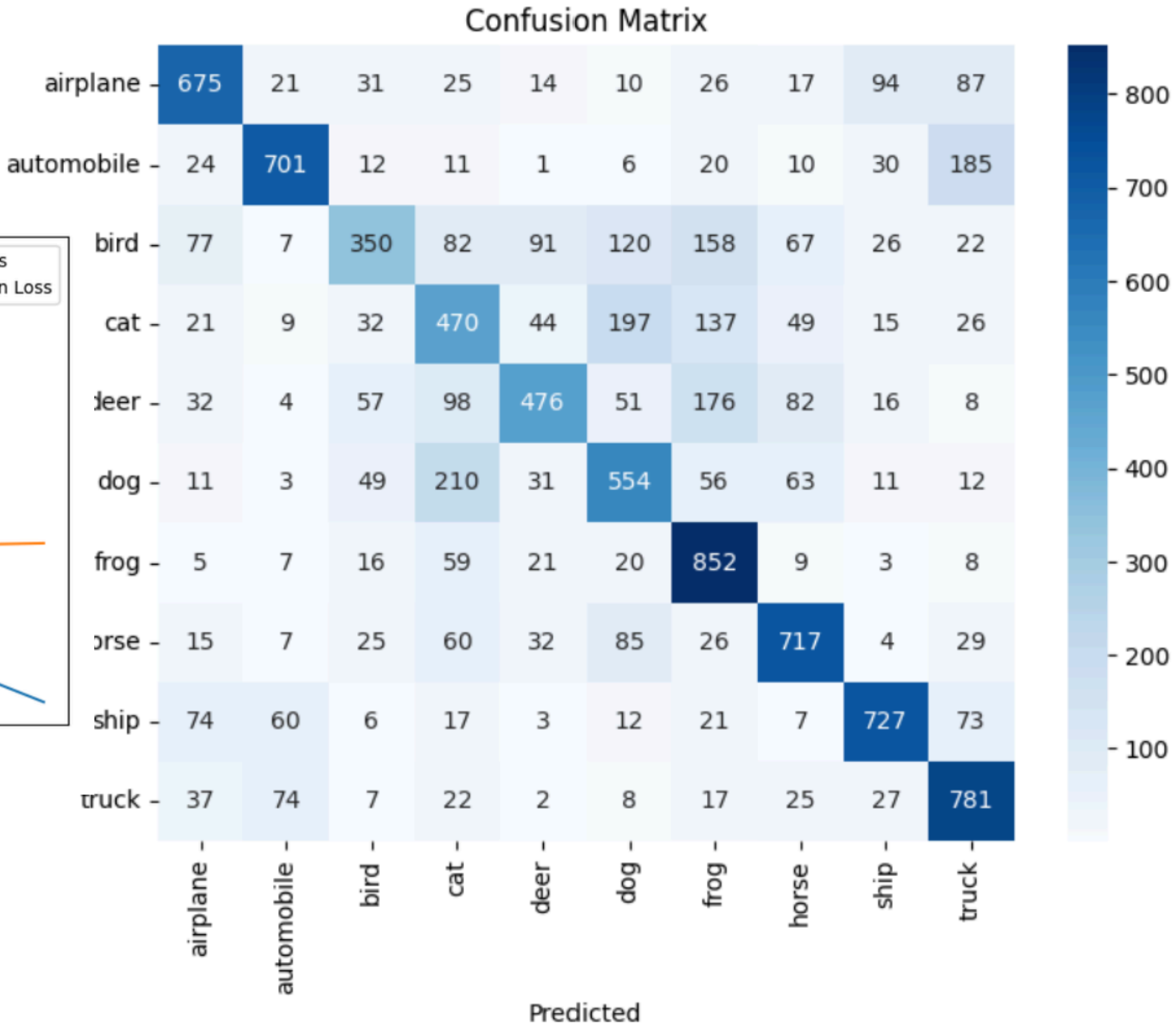
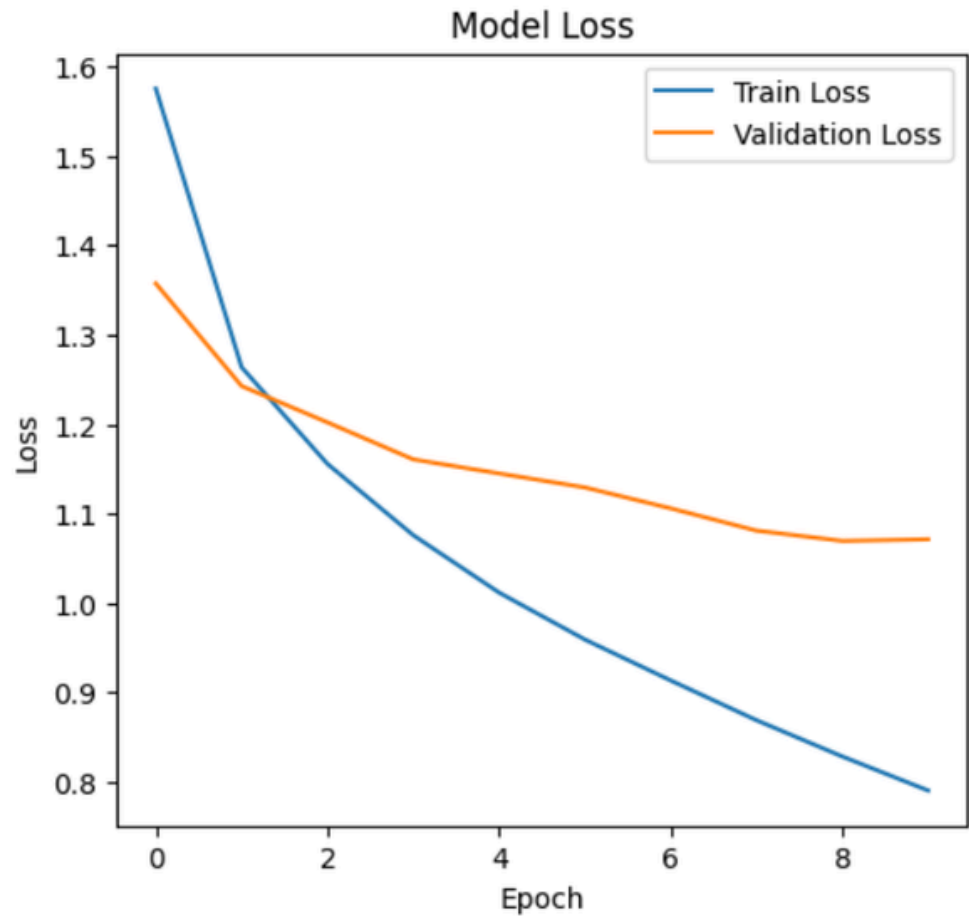
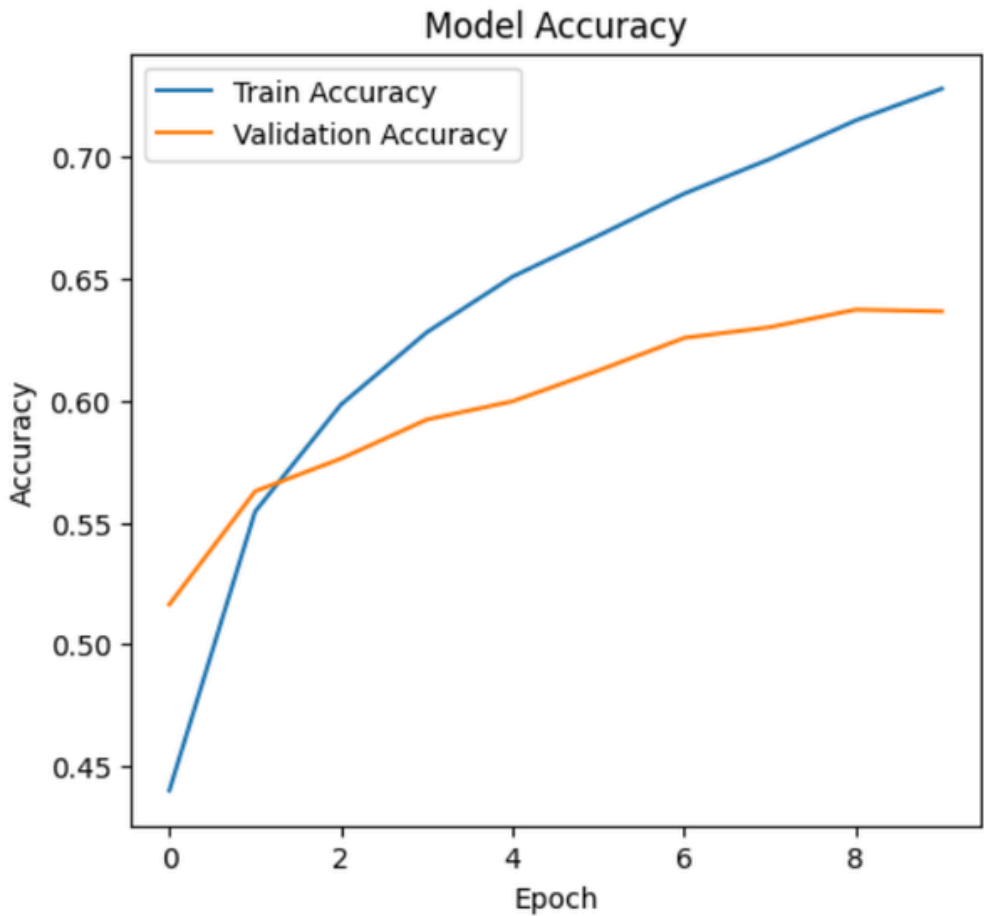
- learning rate = 0.001
- optimizer: adam
- epochs: 10



1-CNN-LAYERED MODEL

Evaluation

Model	Accuracy
1_layeredCNN	0.6319



2-CNN-LAYERED MODEL

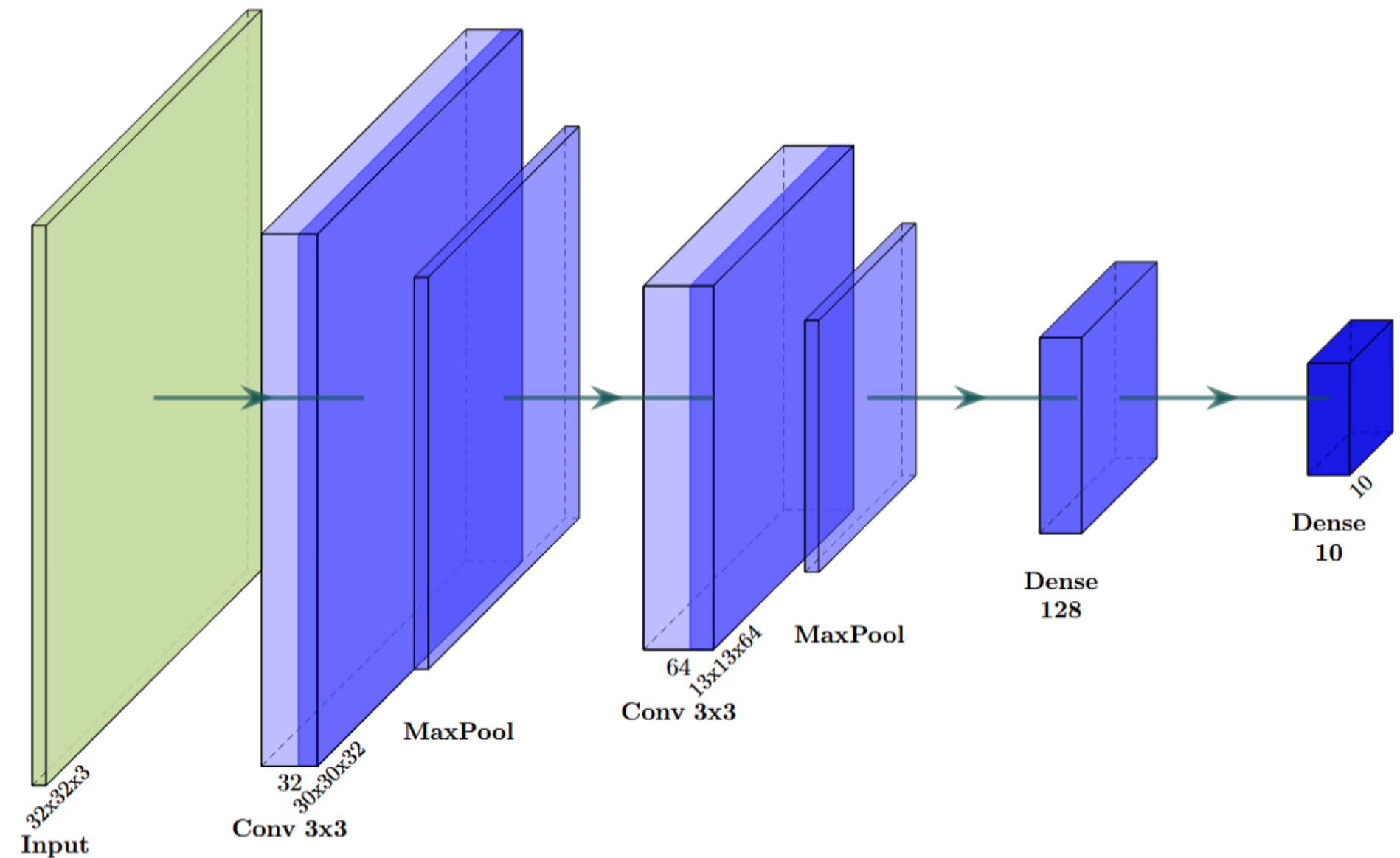
With data augmentation

With regularization:

- BatchNorm
- Dropout with $p=0.5$

Hyperparameters:

- $lr = 10^{-3}$
- optimizer: adam
- early dropping

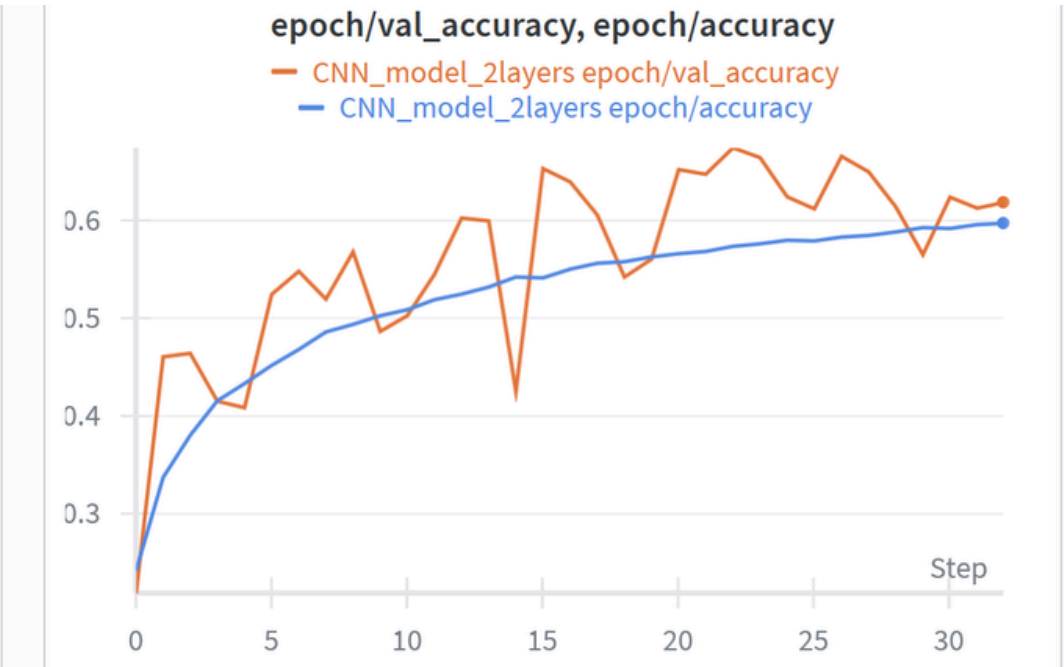
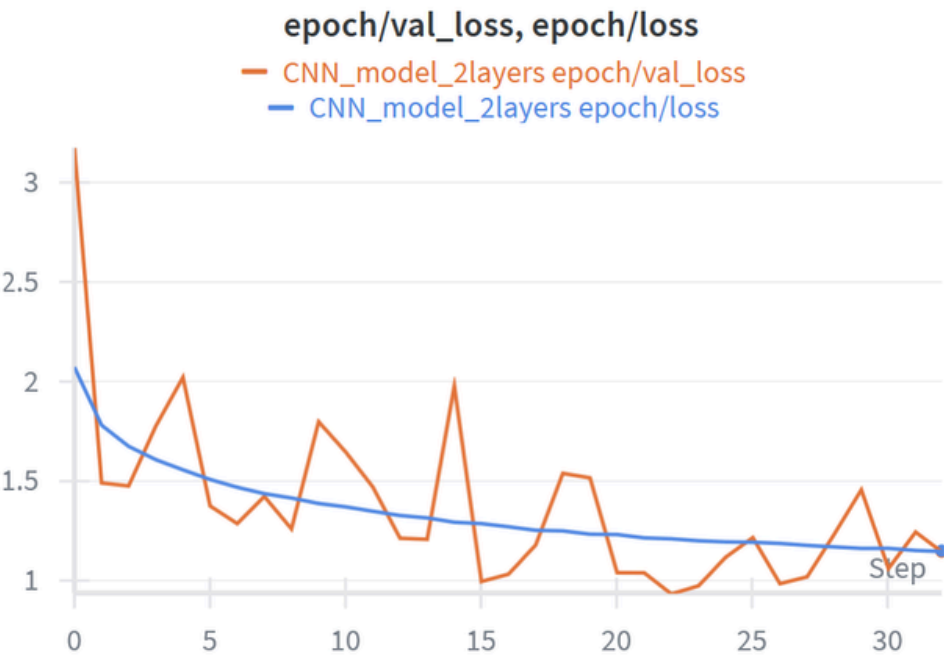
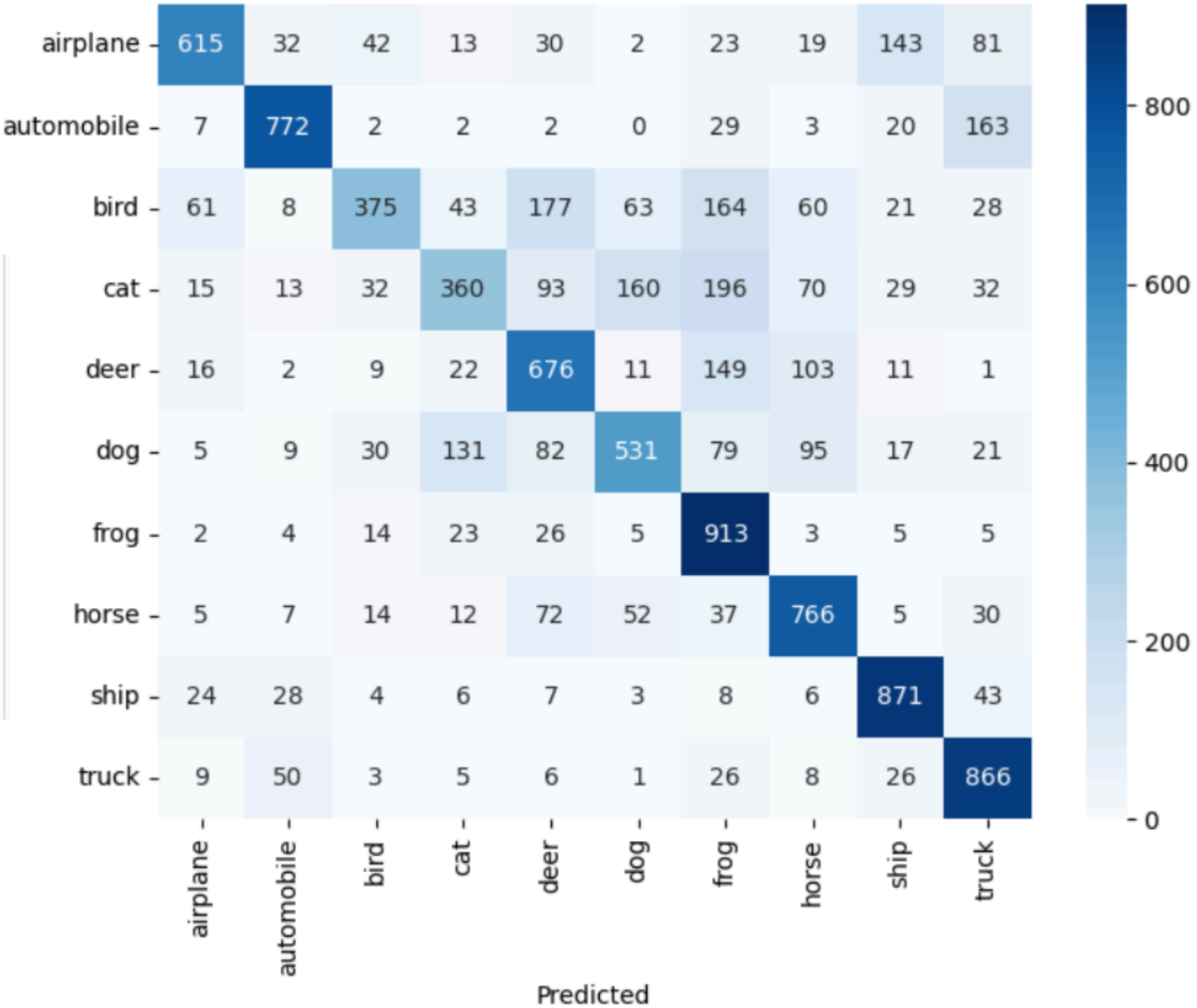


2-CNN-LAYERED MODEL

Evaluation

Model	Accuracy
2_layeredCNN	0.6745

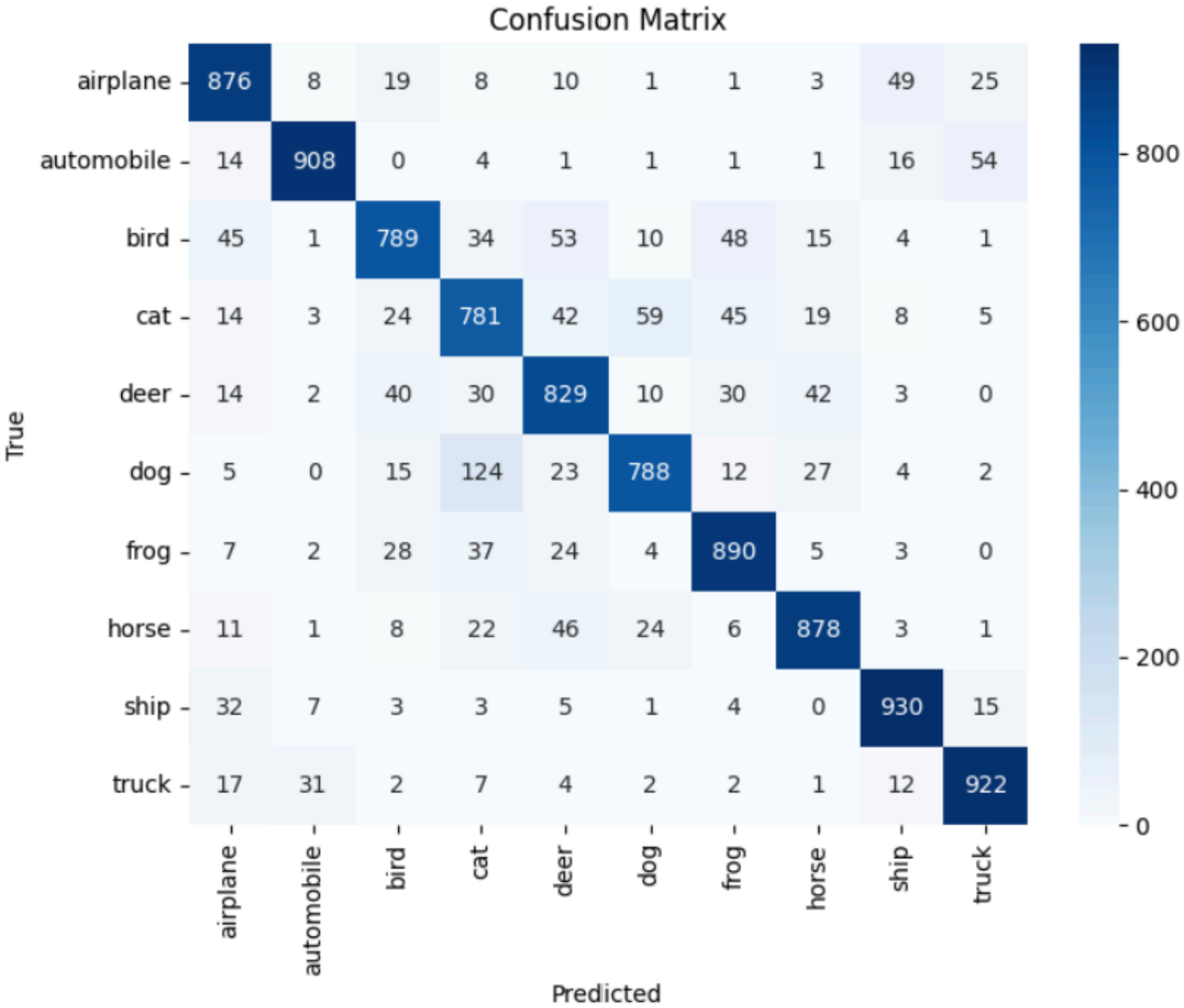
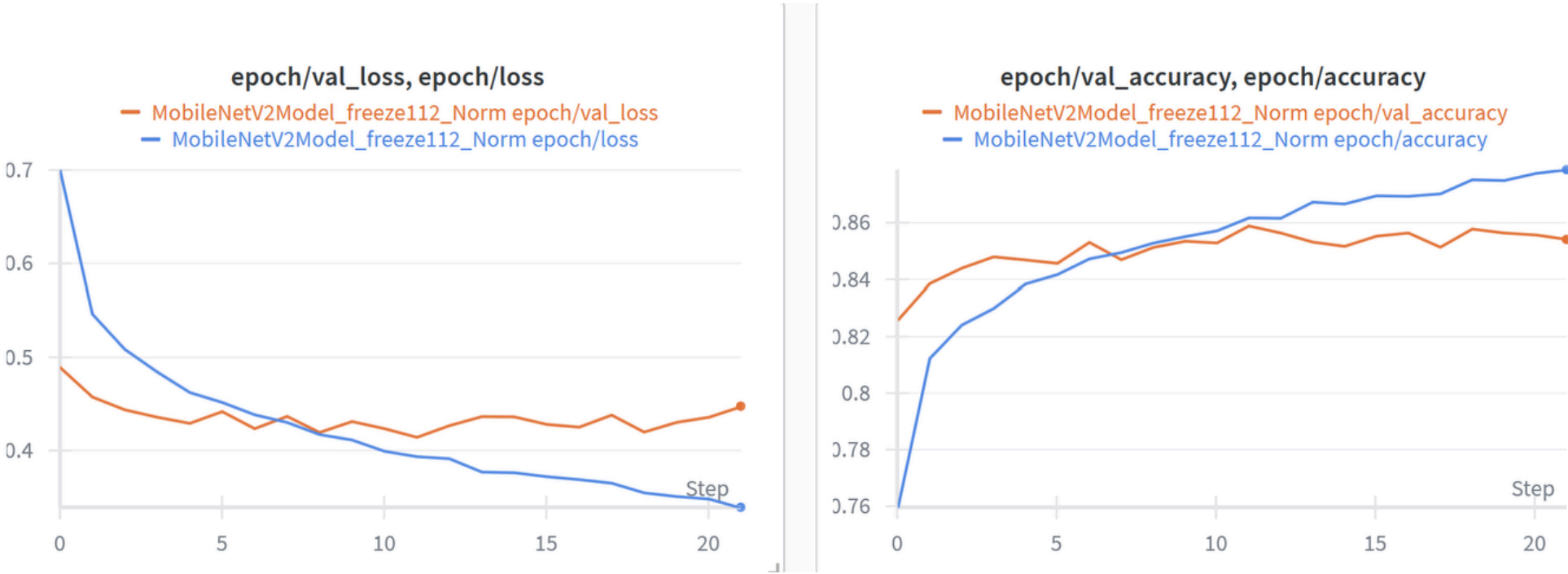
Confusion Matrix



TRANSFER: MOBILENETV2

FREEZE

Model	Accuracy
MobileNetFreeze	0.6745

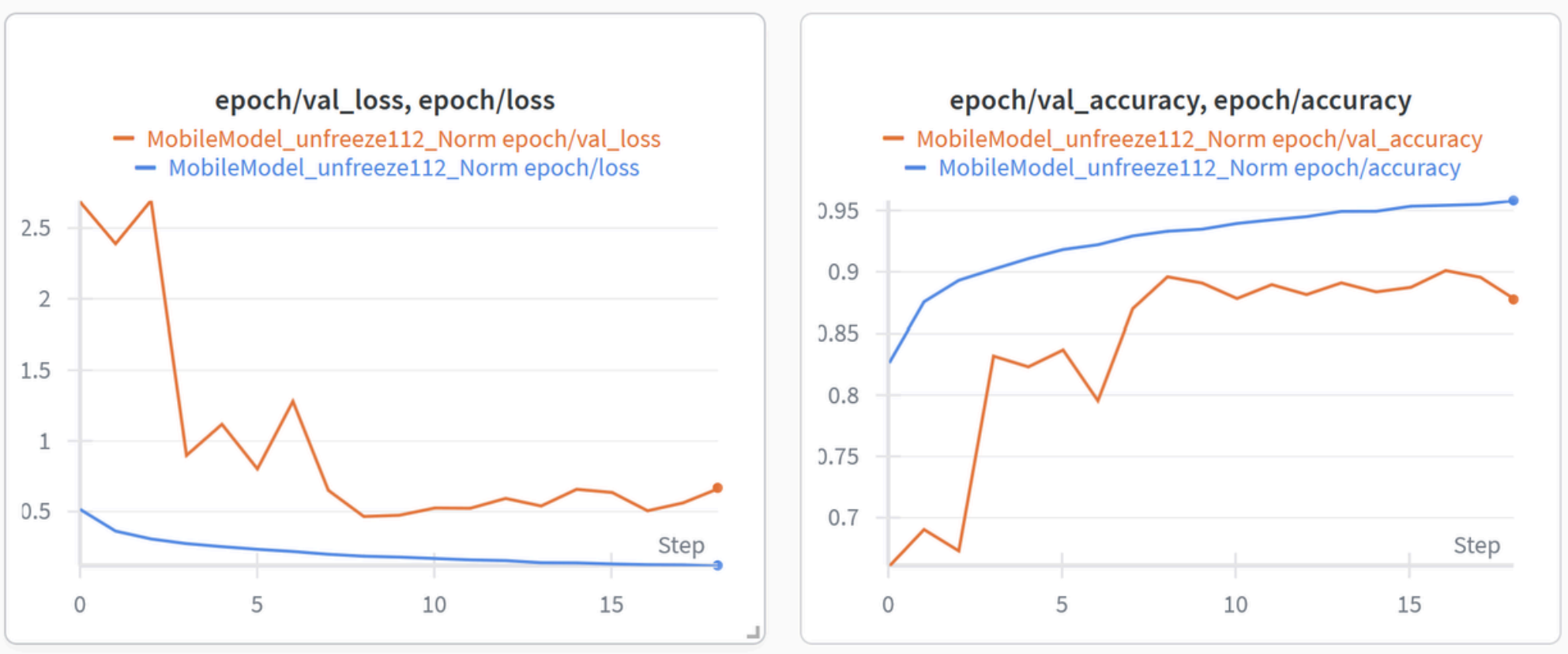
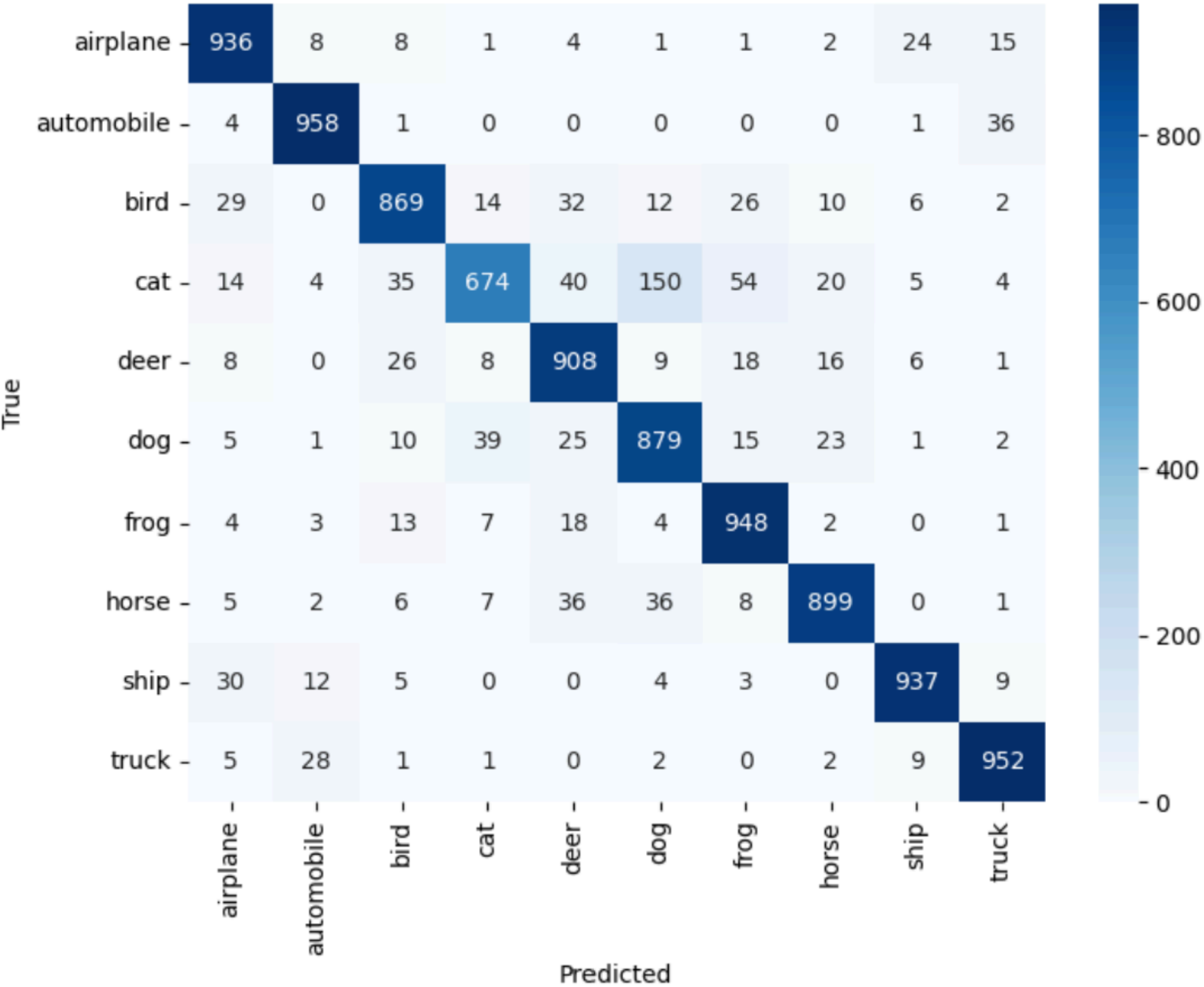


TRANSFER: MOBILENETV2

FINE-TUNING

Model	Accuracy
MobileNetUnFreeze	0.8960

Confusion Matrix



TRANSFER: EFFICIENTNETV2B0

Evaluation

	precision	recall	f1-score	support
airplane	0.93	0.88	0.91	1000
automobile	0.94	0.94	0.94	1000
bird	0.92	0.88	0.90	1000
cat	0.83	0.79	0.81	1000
deer	0.87	0.91	0.89	1000
dog	0.85	0.87	0.86	1000
frog	0.91	0.93	0.92	1000
horse	0.95	0.94	0.94	1000
ship	0.92	0.95	0.93	1000
truck	0.91	0.95	0.93	1000
accuracy			0.90	10000
macro avg	0.90	0.90	0.90	10000
weighted avg	0.90	0.90	0.90	10000

Accuracy: 0.9037
Precision: 0.9038
Recall: 0.9037
F1 Score: 0.9034

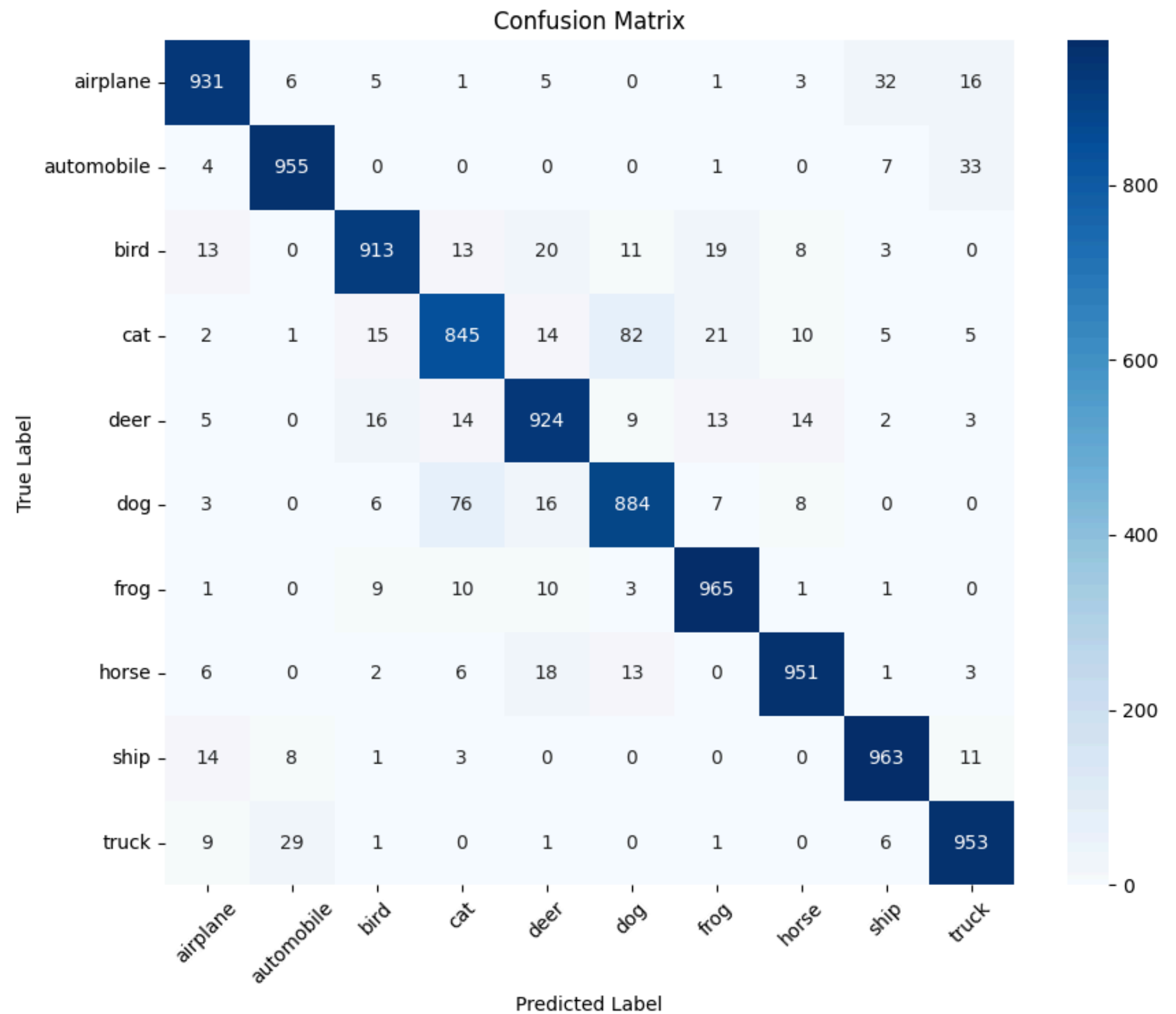


TRANSFER: EFFICIENTNETV2B0

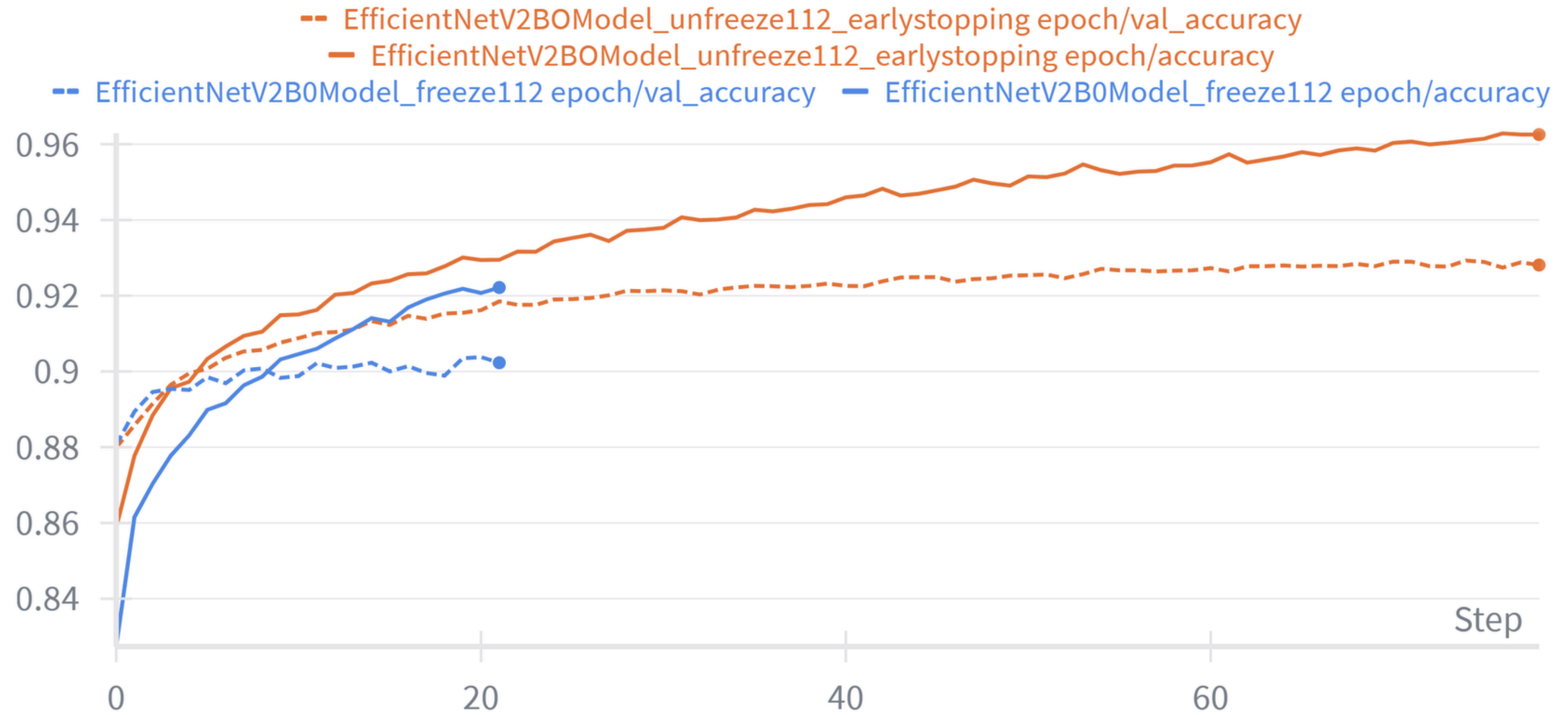
Fine Tuning

	precision	recall	f1-score	support
airplane	0.94	0.93	0.94	1000
automobile	0.96	0.95	0.96	1000
bird	0.94	0.91	0.93	1000
cat	0.87	0.84	0.86	1000
deer	0.92	0.92	0.92	1000
dog	0.88	0.88	0.88	1000
frog	0.94	0.96	0.95	1000
horse	0.96	0.95	0.95	1000
ship	0.94	0.96	0.95	1000
truck	0.93	0.95	0.94	1000
accuracy			0.93	10000
macro avg	0.93	0.93	0.93	10000
weighted avg	0.93	0.93	0.93	10000

Accuracy: 0.9284
Precision: 0.9283
Recall: 0.9284
F1 Score: 0.9282



epoch/val_accuracy, epoch/accuracy

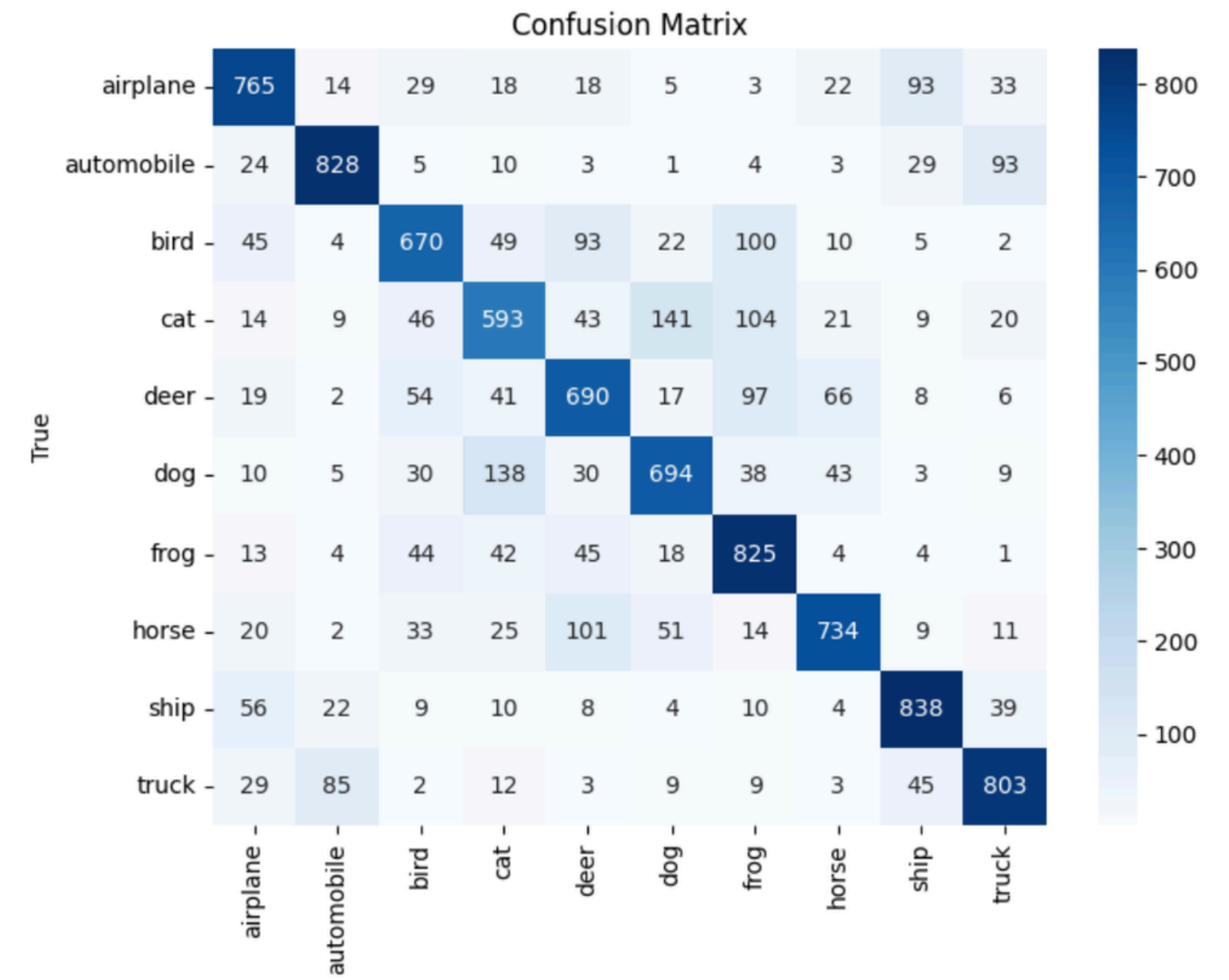


TRANSFER: NASNETMOBILE

Evaluation

Accuracy: 0.7440
Precision: 0.7451
Recall: 0.7440
F1 Score: 0.7434

	precision	recall	f1-score	support
airplane	0.77	0.77	0.77	1000
automobile	0.85	0.83	0.84	1000
bird	0.73	0.67	0.70	1000
cat	0.63	0.59	0.61	1000
deer	0.67	0.69	0.68	1000
dog	0.72	0.69	0.71	1000
frog	0.69	0.82	0.75	1000
horse	0.81	0.73	0.77	1000
ship	0.80	0.84	0.82	1000
truck	0.79	0.80	0.80	1000
accuracy			0.74	10000
macro avg	0.75	0.74	0.74	10000
weighted avg	0.75	0.74	0.74	10000



TRANSFER: DENSENET121

Freeze

Accuracy: 0.5878
Precision: 0.5899
Recall: 0.5878
F1 Score: 0.5848

	precision	recall	f1-score	support
airplane	0.66	0.60	0.63	1000
automobile	0.76	0.59	0.66	1000
bird	0.52	0.55	0.54	1000
cat	0.49	0.34	0.40	1000
deer	0.51	0.57	0.54	1000
dog	0.53	0.52	0.52	1000
frog	0.64	0.67	0.65	1000
horse	0.59	0.59	0.59	1000
ship	0.64	0.72	0.68	1000
truck	0.57	0.74	0.65	1000
accuracy			0.59	10000
macro avg	0.59	0.59	0.58	10000
weighted avg	0.59	0.59	0.58	10000

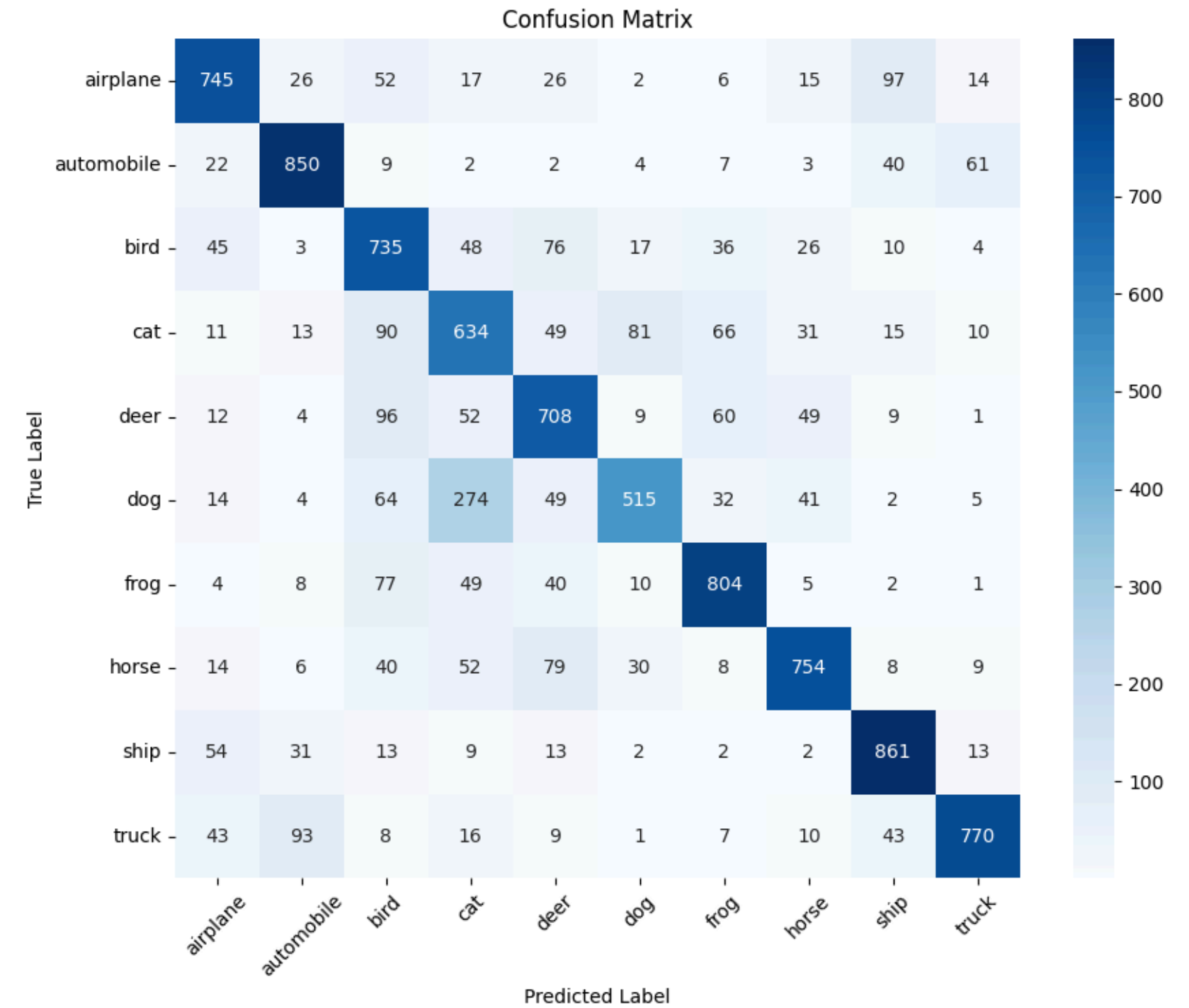


TRANSFER: DENSENET121

Fine-Tuning

Accuracy: 0.7376
Precision: 0.7450
Recall: 0.7376
F1 Score: 0.7374

	precision	recall	f1-score	support
airplane	0.77	0.74	0.76	1000
automobile	0.82	0.85	0.83	1000
bird	0.62	0.73	0.67	1000
cat	0.55	0.63	0.59	1000
deer	0.67	0.71	0.69	1000
dog	0.77	0.52	0.62	1000
frog	0.78	0.80	0.79	1000
horse	0.81	0.75	0.78	1000
ship	0.79	0.86	0.83	1000
truck	0.87	0.77	0.82	1000
accuracy			0.74	10000
macro avg	0.75	0.74	0.74	10000
weighted avg	0.75	0.74	0.74	10000



SUMMARY

Model	Accuracy	Precision	Recall	F1 Score
1_layeredCNN	0.6319	0.6340	0.6319	0.6264
2_layeredCNN	0.6745	0.6840	0.6745	0.6630
MobileNetFreeze	0.6745	0.6840	0.6745	0.6630
MobileNetUnFreeze	0.8960	0.8973	0.8960	0.8946
EfficientNetFreeze	0.9037	0.9038	0.9037	0.9034
EfficientNetUnFreeze	0.9284	0.9283	0.9284	0.9282
NasNETFreeze	0.8607	0.8611	0.8607	0.8604
NasNETUnFreeze	0.8607	0.8611	0.8607	0.8604
DenseNetFreeze	0.5878	0.5899	0.5878	0.5848
DenseNetUnFreeze	0.7376	0.7450	0.7376	0.7374

THANK YOU



LEARNED LESSONS AND CHALLENGES

- Put attention when you use a transfer model what it says about pre-processing the data